

ML-KEM (FIPS 203)

Lecture 17

SEC 595 — Encrypted Computing

- 1 Lattice Reduction and Parameters
- 2 KEMs and the Module Structure
- 3 K-PKE
- 4 Compress and Decompress
- 5 CCA Security and Fujisaki–Okamoto
- 6 The NTT in FIPS 203
- 7 Sizes and Deployment

Reduction takes a bad basis and returns a good basis of the same lattice: subtract multiples of one vector from another, swap when an earlier vector gets shorter, repeat until nothing helps.

Lagrange–Gauss (dimension 2)

Order so $\|b_1\| \leq \|b_2\|$. Set

$$\mu = \frac{\langle b_1, b_2 \rangle}{\langle b_1, b_1 \rangle}, \quad b_2 \leftarrow b_2 - \lfloor \mu \rfloor b_1,$$

then swap and repeat until b_2 stops shrinking.

The shorter the first output vector, the stronger the algorithm.

Root Hermite factor

An algorithm achieves δ_0 if its output has $\|b_1\| = \delta_0^d \cdot \det(\mathcal{L})^{1/d}$. Smaller is stronger.

LLL (1982): polynomial time, $\|b_1\| \leq 2^{(d-1)/2} \lambda_1(\mathcal{L})$; $\delta_0 \approx 1.0219$ in practice.

BKZ- β : LLL plus exact SVP in projected blocks of dimension β . One SVP call costs $2^{0.292\beta}$ classically, $2^{0.265\beta}$ quantumly, and

$$\delta_0(\beta) \approx \left(\frac{\beta}{2\pi e} (\pi\beta)^{1/\beta} \right)^{1/(2(\beta-1))}.$$

$\delta_0(100) \approx 1.00926$ at $2^{29.2}$; $\delta_0(400) \approx 1.00398$ at $2^{116.8}$.

Example — Lagrange–Gauss repairs a bad basis

The lattice $2\mathbb{Z} \times 3\mathbb{Z}$ with bad basis $b_1 = (2, 3)$, $b_2 = (4, 9)$ (defect ≈ 5.918).

- a. Run Lagrange–Gauss: compute μ , $\lfloor \mu \rfloor$ and the new b_2 at each step, and say when it stops.
- b. Check the output against the good basis $(2, 0), (0, 3)$: inner product, determinant, defect.
- c. What is λ_1 ?

The loop

- 1 Fix the scheme: a q -ary lattice (Lecture 5) of dimension d with a planted short vector of norm about $\sigma\sqrt{d}$.
- 2 Check the planted vector is shorter than the Gaussian-heuristic length $\sqrt{d/2\pi e} \det(\mathcal{L})^{1/d}$.
- 3 Find the δ_0 BKZ needs to reach the planted vector.
- 4 Invert $\delta_0(\beta)$; cost $2^{0.292\beta}$.
- 5 If the cost is below 2^λ : raise d (n or k), lower q , or raise σ , and repeat.

Every parameter table in this course is the output of this loop.

Target. 128 bits classical: $0.292\beta \geq 128$, so $\beta = 439$ and

$$\delta_0(439) \approx 1.003732 \quad (\text{LLL: } 1.1892).$$

Lattice. $d = 512$, $\det = q^{256}$, $q = 3329$:

$$\det(\mathcal{L})^{1/d} = \sqrt{3329} \approx 57.70, \quad \text{gh}(\mathcal{L}) \approx 5.475 \cdot 57.70 \approx 315.9.$$

An error of $\sigma = 3.2$ has norm $\approx 3.2\sqrt{512} \approx 72.41$: about 4.36 times shorter than generic.

Published. ML-KEM-512 sits at $\beta \approx 406$, i.e. $0.292 \cdot 406 \approx 118.6$ bits of core-SVP cost.

Coefficient embedding

$$\iota : \mathbb{Z}[X]/(X^n + 1) \rightarrow \mathbb{Z}^n, \quad a_0 + a_1X + \cdots + a_{n-1}X^{n-1} \mapsto (a_0, \dots, a_{n-1}).$$

The image of an ideal is an **ideal lattice**. Coordinatewise, ι sends R^k to $\mathbb{Z}^{nk}$: a rank- k module is a lattice of dimension $d = nk$.

Multiplication by X is the negacyclic shift

$$(a_0, \dots, a_{n-1}) \mapsto (-a_{n-1}, a_0, \dots, a_{n-2}),$$

an isometry. One short vector gives n of the same length.

ML-KEM: $n = 256$, $k \in \{2, 3, 4\}$, so $d = 512, 768, 1024$.

Definition

- $\text{KeyGen}() \rightarrow (ek, dk)$
- $\text{Encaps}(ek) \rightarrow (K, c)$ — takes **no message**
- $\text{Decaps}(dk, c) \rightarrow K'$

Correct if $K' = K$ with overwhelming probability.

The algorithm invents K ; no caller chooses it. For ML-KEM, $K \in \{0, 1\}^{256}$.

	PKE	KEM
Sender supplies	a message m	nothing
Output	c	(K, c)
Receiver learns	m	K
Example	Regev, ElGamal	ML-KEM

This is what TLS needs: a fresh shared key, as Diffie–Hellman gives. ML-KEM replaces DH, not RSA encryption.

No chosen message means no padding to get wrong.

KEM-DEM: use K to key an authenticated cipher (AES-GCM) for the payload.

One ring, three parameter sets

Every parameter set uses

$$n = 256, \quad q = 3329, \quad R_q = \mathbb{Z}_q[X]/(X^{256} + 1).$$

Set	k	η_1	η_2	(d_u, d_v)	Cat.
ML-KEM-512	2	3	2	(10, 4)	1
ML-KEM-768	3	2	2	(10, 4)	3
ML-KEM-1024	4	2	2	(11, 5)	5

Only the module rank k really varies. $3329 < 2^{12}$, so a ring element is $256 \cdot 12/8 = 384$ bytes.

Categories 1, 3, 5: as hard as AES-128, -192, -256.

$$\mathbf{A} \in R_q^{k \times k}, \quad \mathbf{s}, \mathbf{e} \in R_q^k, \quad \mathbf{t} = \mathbf{A}\mathbf{s} + \mathbf{e}.$$

- $\mathbf{A}$ uniform, expanded from a 32-byte seed ρ by SHAKE128 — never transmitted.
- Coefficients of $\mathbf{s}, \mathbf{e}$ from CBD_{η_1} (Lecture 6), so $\|\mathbf{s}\|_\infty, \|\mathbf{e}\|_\infty \leq \eta_1 \leq 3$.

$(\mathbf{A}, \mathbf{t})$ is exactly an MLWE sample (Lecture 6). Without $\mathbf{e}$, $\mathbf{s} = \mathbf{A}^{-1}\mathbf{t}$ by linear algebra.

- ① **One NTT.** Changing n changes the two-adicity q needs, hence q and the whole transform. Changing k changes a loop bound.
- ② **Granularity.** With power-of-two n the only move is $256 \rightarrow 512$. k moves in steps of 256 lattice dimensions.
- ③ **Less structure.** A rank- k module has fewer automorphisms than a ring of degree nk — a hedge against structural attacks on RLWE.

Example — a small Module-LWE instance

$\mathbb{Z}_{17}[X]/(X^4 + 1)$, $k = 2$; write (a_0, a_1, a_2, a_3) for $a_0 + a_1X + a_2X^2 + a_3X^3$.

$$\mathbf{A} = \begin{pmatrix} (3, 1, 4, 1) & (5, 9, 2, 6) \\ (2, 7, 1, 8) & (1, 4, 1, 5) \end{pmatrix},$$

$$\mathbf{s} = \begin{pmatrix} (1, 0, -1, 1) \\ (0, 1, 1, 0) \end{pmatrix}, \quad \mathbf{e} = \begin{pmatrix} (0, 1, 0, -1) \\ (1, 0, 0, 1) \end{pmatrix}.$$

- a. Compute $\mathbf{t} = \mathbf{A}\mathbf{s} + \mathbf{e}$ in centered representatives.
- b. Which of $\mathbf{s}$, $\mathbf{e}$, $\mathbf{t}$ looks uniform?

Inside ML-KEM is an IND-CPA encryption scheme for 256-bit messages, **K-PKE**. It is never exposed on its own.

KeyGen

- 1 Seed d ; $(\rho, \sigma) = G(d\|k)$, $G = \text{SHA3-512}$.
- 2 Expand $\mathbf{A}$ from ρ .
- 3 $\mathbf{s}, \mathbf{e} \leftarrow \text{CBD}_{\eta_1}^k$ from σ .
- 4 $\mathbf{t} = \mathbf{A}\mathbf{s} + \mathbf{e}$.
- 5 $ek_{\text{PKE}} = (\mathbf{t}, \rho)$, $dk_{\text{PKE}} = \mathbf{s}$.

Encrypt(ek_{PKE}, m, r), $m \in \{0, 1\}^{256}$

$\mathbf{y} \leftarrow \text{CBD}_{\eta_1}^k$, $\mathbf{e}_1 \leftarrow \text{CBD}_{\eta_2}^k$, $e_2 \leftarrow \text{CBD}_{\eta_2}$, all from r .

$$\mathbf{u} = \mathbf{A}^\top \mathbf{y} + \mathbf{e}_1, \quad v = \mathbf{t}^\top \mathbf{y} + e_2 + \lfloor q/2 \rfloor m.$$

Output $c = (\text{Compress}_{d_u}(\mathbf{u}), \text{Compress}_{d_v}(v))$.

Decrypt($\mathbf{s}, c$)

Decompress to $\mathbf{u}', v'$; $w = v' - \mathbf{s}^\top \mathbf{u}'$; output $\text{Compress}_1(w)$: bit 1 if a coefficient is nearer $q/2$ than 0.

$\lfloor q/2 \rfloor = 1665$: the Δ of Lecture 1 with $t = 2$.

Proposition

With compression errors $\mathbf{u}' = \mathbf{u} + \mathbf{c}_u$, $v' = v + c_v$,

$$w = \lfloor q/2 \rfloor m + \delta, \quad \delta = \mathbf{e}^\top \mathbf{y} - \mathbf{s}^\top \mathbf{e}_1 + e_2 + (c_v - \mathbf{s}^\top \mathbf{c}_u),$$

and $\text{Compress}_1(w) = m$ whenever $\|\delta\|_\infty < q/4$.

Proof. $\mathbf{t}^\top \mathbf{y} = \mathbf{s}^\top \mathbf{A}^\top \mathbf{y} + \mathbf{e}^\top \mathbf{y}$ and $\mathbf{s}^\top \mathbf{u} = \mathbf{s}^\top \mathbf{A}^\top \mathbf{y} + \mathbf{s}^\top \mathbf{e}_1$; the $\mathbf{s}^\top \mathbf{A}^\top \mathbf{y}$ terms cancel.

Diffie–Hellman, additively: the sender never knows $\mathbf{s}$, the receiver never knows $\mathbf{y}$, both reach $\mathbf{s}^\top \mathbf{A}^\top \mathbf{y}$.

The template of Lecture 10, with threshold $q/4$ (one bit per coefficient).

The error terms

- $\mathbf{e}_1$ hides $\mathbf{A}^\top \mathbf{y}$. It enters δ multiplied by $\mathbf{s}$: $\|\mathbf{s}^\top \mathbf{e}_1\|_\infty \leq kn\eta_1\eta_2$.
- e_2 hides $\mathbf{t}^\top \mathbf{y}$. It enters unmultiplied: at most η_2 . Nearly free.
- $\mathbf{e}$ from KeyGen returns as $\mathbf{e}^\top \mathbf{y}$ in every ciphertext.

Worst case, ML-KEM-768, no compression

$$kn\eta_1^2 + kn\eta_1\eta_2 + \eta_2 = 3072 + 3072 + 2 = 6146 > q/4 = 832.25.$$

The worst case fails by a factor over seven. It is absurdly pessimistic: the true tail is Gaussian.

FIPS 203 computes the failure probability exactly:

	ML-KEM-512	ML-KEM-768	ML-KEM-1024
failure	2^{-139}	2^{-164}	2^{-174}

Acceptable for a KEM: far below any number of protocol runs, and a failure returns a pseudorandom key, not a visible error.

Not for FHE: a failure silently returns a wrong plaintext in the middle of a computation, so BFV and BGV make correctness deterministic.

Still must be below about $2^{-\lambda}$: failing ciphertexts correlate with **s** and enable reaction attacks.

Example — K-PKE end to end

$n = 4$, $q = 17$, $k = 2$, no compression, keys of the previous example.
 $m = (1, 0, 1, 1)$ and

$$\mathbf{y} = \begin{pmatrix} (1, 1, 0, 0) \\ (0, -1, 1, 0) \end{pmatrix}, \quad \mathbf{e}_1 = \begin{pmatrix} (0, 0, 1, 0) \\ (1, 0, 0, 0) \end{pmatrix}, \quad \mathbf{e}_2 = (0, 1, 0, -1).$$

- a. Compute $\mathbf{u}$ and v .
- b. Compute $w = v - \mathbf{s}^\top \mathbf{u}$ and δ .
- c. Check $\delta = \mathbf{e}^\top \mathbf{y} - \mathbf{s}^\top \mathbf{e}_1 + \mathbf{e}_2$ and $\|\delta\|_\infty < q/4$; decrypt.

Definition

For $d < \lceil \log_2 q \rceil$,

$$\text{Compress}_d(x) = \left\lfloor \frac{2^d}{q} x \right\rfloor \bmod 2^d, \quad \text{Decompress}_d(y) = \left\lfloor \frac{q}{2^d} y \right\rfloor,$$

applied coefficientwise.

Rounding (Lecture 1) keeps the error centered at zero, so it does not accumulate across $\mathbf{s}^\top \mathbf{c}_u$.

Error

$$|\text{Decompress}_d(\text{Compress}_d(x)) - x| \leq B_d = \lfloor q/2^{d+1} \rfloor \text{ (centered mod } q\text{)}.$$

Why d_u is big and d_v is small

d	B_d	Used for
1	832	the message: $\text{Decompress}_1(m) = 1665 m$
4	104	d_v , ML-KEM-512/768
5	52	d_v , ML-KEM-1024
10	2	d_u , ML-KEM-512/768
11	1	d_u , ML-KEM-1024

c_u is multiplied by s (factor kn): compress gently.

c_v enters unmultiplied: $104 \ll 832.25$, compress hard.

Same asymmetry as e_1 against e_2 .

The ciphertext size

d bits per coefficient: $256d/8 = 32d$ bytes per ring element.

Proposition

$$|c| = \underbrace{32 d_u k}_u + \underbrace{32 d_v}_v = 32 (d_u k + d_v) \text{ bytes.}$$

ML-KEM-768: 1536 bytes uncompressed $\rightarrow$ 1088, a 29% saving.

Compression changes the scheme: its errors appear in δ , and the security analysis is stated for the compressed ciphertext.

Add $\text{Decompress}_1(m')$ to v : the ciphertext now decrypts to $m \oplus m'$.

With a decryption oracle, one query recovers any challenge.

The threat is real: a TLS server decapsulates whatever it is sent and behaves observably afterwards. Bleichenbacher (1998) broke RSA PKCS#1 v1.5 with only a valid/invalid padding signal.

Homomorphic schemes are malleable and cannot be CCA2 secure (Lecture 8). ML-KEM gives up malleability to get CCA2.

Attacker		Challenger
	$\longleftarrow$	$(ek, dk) \leftarrow \text{KeyGen}; \text{ send } ek$
	$\longleftarrow$	$(K_0, c^*) \leftarrow \text{Encaps}(ek), K_1 \text{ uniform}$
	$\longleftarrow$	random bit b ; send (c^*, K_b)
query $c \neq c^*$	$\longleftrightarrow$	answer $\text{Decaps}(dk, c)$
guess b'	$\longrightarrow$	

IND-CCA2

For every efficient attacker, $\left| \Pr[b' = b] - \frac{1}{2} \right| \leq \text{negl}(\lambda)$.

Decapsulation queries are allowed before and after the challenge. No chosen messages: the attacker only tells real K from random.

The Fujisaki–Okamoto transform

Turns an IND-CPA PKE into an IND-CCA2 KEM, in the random oracle model.

- 1 **Derandomise.** Derive the encryption randomness from the message: $r = G(m)$. Anyone holding m can recompute the unique ciphertext.
- 2 **Re-encrypt on decryption.** Decrypt to m' , re-encrypt with $G(m')$, compare with the received ciphertext. Any modification changes m' , hence the re-encryption, and is caught.

Random oracle model

A hash is modelled as a truly random function; the proof may see every query. No real hash is one, so the proof is evidence, not a theorem about the deployed system.

$H = \text{SHA3-256}$, $G = \text{SHA3-512}$, $J = \text{SHAKE256}$.

KeyGen

K-PKE.KeyGen, plus a 32-byte **rejection key** z .

$$ek = ek_{\text{PKE}}, \quad dk = dk_{\text{PKE}} \parallel ek_{\text{PKE}} \parallel H(ek_{\text{PKE}}) \parallel z.$$

Encaps(ek)

- 1 $m \leftarrow \{0, 1\}^{256}$.
- 2 $(K, r) = G(m \parallel H(ek))$.
- 3 $c = \text{K-PKE.Encrypt}(ek, m, r)$; output (K, c) .

One hash call gives both K and r . $H(ek)$ binds K to the key.

Decaps(dk, c)

- 1 $m' = \text{K-PKE.Decrypt}(dk_{\text{PKE}}, c)$.
- 2 $(K', r') = G(m' \| H(ek_{\text{PKE}}))$.
- 3 $\tilde{K} = J(z \| c)$.
- 4 $c' = \text{K-PKE.Encrypt}(ek_{\text{PKE}}, m', r')$.
- 5 If $c' \neq c$, set $K' = \tilde{K}$.
- 6 Output K' .

Decaps runs a full encryption, so it is slower than Encaps.

dk stores ek_{PKE} , $H(ek_{\text{PKE}})$ and z only so that this check needs no other input.

Step 5 returns $\bar{K} = J(z\|c)$, never an error.

- **No oracle.** A rejected ciphertext gives a key that looks valid; the protocol fails later, identically for every bad input.
- **Deterministic.** $\bar{K}$ depends only on z and c .
- **Absorbs decryption failures.** A rare failure ($m' \neq m$) also returns $\bar{K}$; the attacker cannot tell why.

Requirement

The comparison $c' \neq c$ must be **constant time**. An early-exit compare reintroduces the oracle.

The proof reduces IND-CCA2 of ML-KEM to IND-CPA of K-PKE (ROM); its loss grows with the failure probability.

The incomplete NTT

A negacyclic NTT of length 256 needs a primitive 512-th root.

$$q - 1 = 3328 = 2^8 \cdot 13.$$

$256 \mid q - 1$ ($\zeta = 17$ has order 256), but $512 \nmid q - 1$. The transform runs 7 layers and stops:

$$\mathbb{Z}_{3329}[X]/(X^{256} + 1) \cong \prod_{i=0}^{127} \mathbb{Z}_{3329}[X]/(X^2 - \zeta^{2 \operatorname{brv}(i)+1}).$$

128 quadratic factors, not 256 linear ones. The “pointwise” product is `BaseCaseMultiply`: degree-1 polynomials mod $X^2 - \zeta_i$.

FIPS 203 works in the NTT domain: $\hat{\mathbf{A}}$ is sampled there and ek stores $\hat{\mathbf{t}}$.

Example — counting the layers

ML-KEM uses $q = 3329$; ML-DSA (Lecture 18) uses $q = 8380417$.

- a. Factor $q - 1$ for each and read off the two-adicity.
- b. How many layers can a negacyclic NTT of length 256 run?
- c. What is the degree of the resulting factors?
- d. What does one pointwise step cost, in coefficient multiplications?

Three formulas

$$ek = 384k + 32, \quad dk = 768k + 96, \quad c = 32(d_u k + d_v).$$

ek : $\hat{\mathbf{t}}$ ($384k$) and ρ (32) — not $384k^2$ for $\mathbf{A}$.

dk : $\hat{\mathbf{s}}$ ($384k$), ek ($384k + 32$), $H(ek)$ (32), z (32).

Set	ek	dk	c
ML-KEM-512	800	1632	768
ML-KEM-768	1184	2400	1088
ML-KEM-1024	1568	3168	1568

Example — derive the sizes

$$(k, d_u, d_v) = (2, 10, 4), (3, 10, 4), (4, 11, 5).$$

- a. Derive ek , dk and c for all three sets.
- b. For ML-KEM-768, how many bytes of c are $\mathbf{u}$, and how many v ?
- c. Which size would change, and by how much, if $\mathbf{A}$ were sent instead of ρ ?

TLS 1.3: the client's `key_share` is ek , the server's is c . Deployed as a hybrid with X25519:

Message	Contents	Bytes
Client	ek (1184) + X25519 (32)	1216
Server	c (1088) + X25519 (32)	1120
Total	(X25519 alone: 64)	2336

The two shared secrets are concatenated into the key schedule: secure if *either* half holds.

Why: harvest-now-decrypt-later, and the 2022 breaks of Rainbow and SIKE (Lecture 16). ML-KEM gives confidentiality only; authentication needs ML-DSA.

- 1 Parameters come from lattice reduction: choose them so the BKZ block size needed costs more than 2^λ .
- 2 ML-KEM: $n = 256$, $q = 3329$, rank $k \in \{2, 3, 4\}$; the public key $\mathbf{t} = \mathbf{A}\mathbf{s} + \mathbf{e}$ is an MLWE instance.
- 3 K-PKE decrypts to $\lfloor q/2 \rfloor m + \delta$, correct when $\|\delta\|_\infty < q/4$; failure $\leq 2^{-139}$, fine for a KEM, not for FHE.
- 4 Compression sets $|c| = 32(d_u k + d_v)$; d_u large because $\mathbf{c}_u$ meets $\mathbf{s}$.
- 5 FO with implicit rejection lifts IND-CPA to IND-CCA2; the re-encryption check must be constant time.
- 6 $q - 1 = 2^8 \cdot 13$: the NTT stops at 128 quadratic factors.